

SMART QUEUE MANAGEMENT SYSTEM FOR RATION SHOPS

1. Problem Statement

Traditional ration shops rely on manual token distribution and physical queues for distributing essential commodities. This often results in long waiting times, overcrowding, lack of transparency, and inconvenience for ration card holders. Customers are unable to know their queue status or expected waiting time, while shop administrators face challenges in managing queues efficiently. Therefore, a Smart Queue Management System for Ration Shop is required to provide online token booking, real-time queue tracking, and efficient queue management, improving service quality and reducing waiting time.

2. Objectives

Primary Objective

To develop a web-based Smart Queue Management System for ration shops that reduces waiting time and improves queue management.

Specific Objectives

To provide secure user registration and login.

To enable online token booking for ration card holders.

To display real-time queue status.

To show estimated waiting time.

To provide an admin dashboard for queue management.

To reduce overcrowding at ration shops.

To maintain queue history and reports.

3. Scope

The Smart Queue Management System is designed specifically for government ration shops. It allows ration card holders to book tokens online, track their queue status, and receive estimated waiting time. The system helps administrators manage queues efficiently, reduce manual work, and improve the distribution of essential commodities such as rice, wheat, sugar, and kerosene. Future enhancements may include QR code-based tokens, SMS notifications, and a mobile application.

4. Expected Outcomes

Reduced waiting time at ration shops.

Online token booking for ration card holders.

Real-time queue tracking.

Reduced overcrowding.

Better transparency in ration distribution.

Efficient queue management by administrators.

Improved customer satisfaction.